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1. Introduction to Web Accessibility

1.1. Objectives

To get introduction on:

- Web Accessibility & The benefits to the various types of users
- Guidelines to build Web accessibility using HTML, CSS, JavaScript
- Initiatives towards it from governments across the world
- References for additional details

1.2.Web Accessibility

Web Accessibility means: *Web for People with disabilities*; Refers to the 'inclusive practice' of removing barriers that prevent access to websites by people with disabilities. People with disabilities also need to perceive, understand, navigate, and interact with the Web, and they can also contribute to the Web. Web accessibility also benefits others, including older people with changing abilities due to aging or temporary disablements.

The World Wide Web has rapidly become the dominant Internet tool, combining hypertext and multimedia to provide a network of educational, governmental, and commercial resources. Because of the multimedia nature of the web and the poor design of some websites, many Internet users cannot access the full range of resources Cannot see graphics, Cannot hear audio Cannot operate a mouse, Use slow Internet connections and modems or equipment that cannot easily download large files. Have difficulty navigating sites that are poorly organized with unclear directions because they have learning disabilities, speak English as a second language, or are younger than the average user.

1.3.Benefits to the Users

To enable the usage of the powerful Web features, 'Web Accessibility' has been initiated by the various organizations, including government agencies, private agencies, etc.

Additionally the Web Accessibility provides benefits to all kinds of users in various manners:

- The Web hosts many important resources in every day aspects of life: Education, Employment, Government, Commerce, Health Care, Recreation, and more. So it is essential that the Web be accessible in order to provide **equal access** and **equal opportunity** to people with disabilities also.

- Accessible Web sites help people with disabilities to more actively **participate in society**.

- Web Accessible sites provide **unprecedented access to information and interactions** for many people with disabilities.

- Web accessibility, being an aspect of corporate social responsibility, helps the organizations to provide the support back to the community

- Web accessibility is becoming part of laws and policies in some countries to help the society and organization.

2. Guidelines for Web Accessibility

When a web site is being designed and developed, all users must have equal access to information and functionality, irrespective of the technology used.

The sites shall include addressing the needs of the following users as well:

Visual: Visual impairments, includes poor eyesight, low vision, blindness, various color blindness.

Motor/Mobility: Inability to use hands or loss of muscle control

Auditory: Hearing impairments/deafness

Seizures: Photo-epileptic seizures caused by visual stroke or flashing effects

Cognitive/Intellectual: Developmental, learning and cognitive disabilities, affecting memory, attention, developments, problem solving and logic skills.

There are many guidelines defined and being followed in developing the Web Accessible sites with respect to the standard User Interface Technologies, such as HTML, CSS and JavaScript. The following sub-sections shall provide detailed descriptions of developing Web Accessible web sites using these technologies.

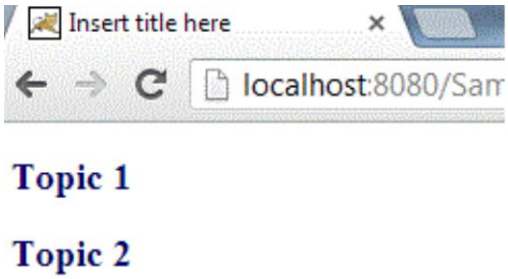
2.1.HTML Guidelines for Web Accessibility

The section provides HTML elements that will be useful to develop Web Accessible sites.

2.1.1. HTML Web Accessibility Elements

Inaccessible Use of < font size > tag for headers

In this example, the **<font size="+1" color="#000066" (navy) >** and **<b >** tags have been used to create large sub headings.

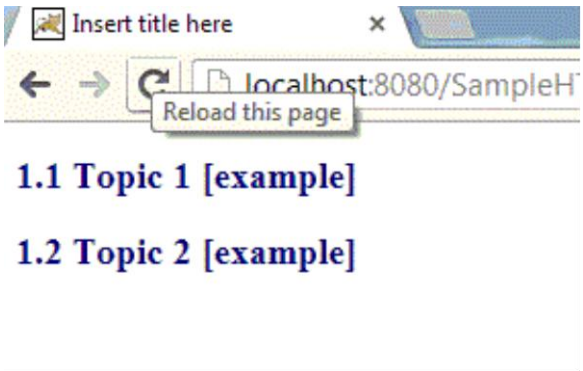


A screen reader set to a scanning mode would not list these as topics. In addition the code contains more formatting tags than needed.

View the Code : `<p> <font size="+1" color="#000066"> <b> Topic 1 </b> </font> </p>`

Accessible Use of H2 tag for headers

In this example the H2 tag has been used and has been styled so that it is automatically navy.



A screen reader set to a scanning mode would list "Topic 1" then "Topic 2".

View the Code

```
h2 {color: #000066}  …  <body>  <h2> Topic 1 </h2>
```

Tagging Large Sizes Appropriately: Do not use the H tags just to format text to a larger size.

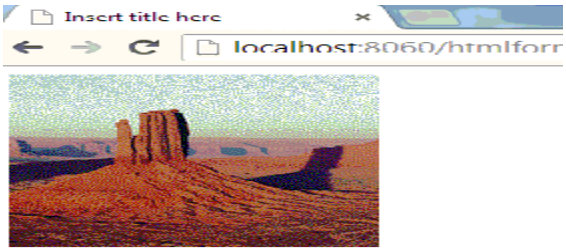
2.1.2. HTML Image ALT Tag

Note: The term "ALT tag" is a common shorthand term used to refer to the ALT attribute within in the IMG tag.

Implementing Alt Text

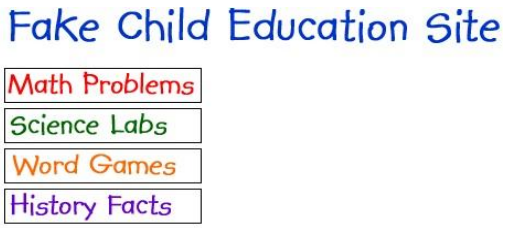
The ALT tag adds a text description to an image on a Web page, and should be used for all images, graphical bullets, and graphical horizontal rules.

Sample ALT Text Code <img src="Desert.gif" alt="Desert logo">



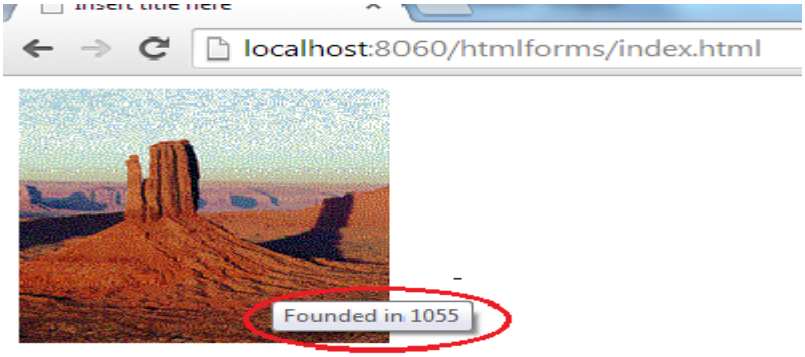
Screen Reader Output With ALT Tag

```
<img src="K12Title.gif" alt="Fake Child Education Site Label">
</p>
<p> <img src="K12MathProblems.gif" alt="Math Problems">
</p><p> <img src="K12MathProblems.gif" alt="Math Problems"
<br><p> <img src="K12MathProblems.gif" alt="Math Problems"
<br><p> <img src="K12MathProblems.gif" alt="Math Problems"
<br>
```



TITLE Tooltip Example HTML Code

```
<img src="Desert.gif" alt="Desert State University"
title="Founded in 1055" >
```



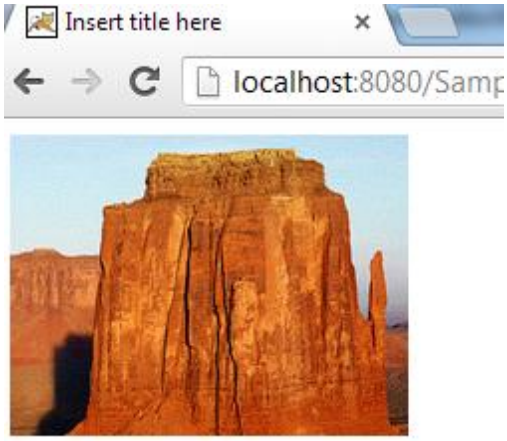
Example: (Hold cursor over image to test tooltip)

2.1.3. HTML Inaccessible ALT Tag for Graphic Link

This code is meant to implement the State logo as a link to the homepage, but isn't fully accessible.

View the Code

```
<a href="http://www.gov.edu" > <img src="govlogo.gif" alt="State Logo"> </a>
```



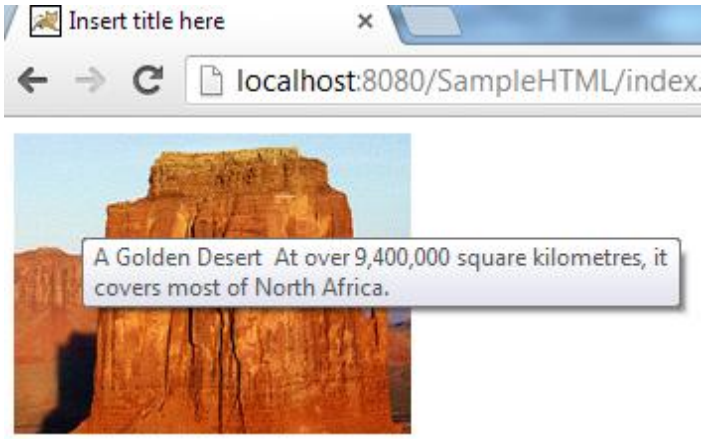
A screen reader would say "State Logo," but would not indicate that this link goes to the State homepage.

<u>Accessible ALT Tag for Graphic Link</u>

This code is a more accessible implementation of the State logo as a link to the homepage.

View the Code

<img src="sample.jpg" width="200" height="150" alt="A Golden Desert At over 9,400,000 square kilometres, it covers most of North Africa.">



The screen reader would say "State homepage," which indicates the link's destination.

2.1.4. HTML ALT Text for Filler or Decorative Images

The ALT tag for these images should be blank (<alt= " " >) so that screen readers will ignore them altogether.. The example below will use a sample image of a rainbow toolbar, followed by some accessible and inaccessible code examples.

Accessible Toolbar Image Code:

<img src="examples/tooltip.gif" alt=" " >

Less Usable Toolbar Image Code

<img src="examples/tooltip.gif" alt="A Rainbow line used as toolbar" >

Inaccessible Toolbar Image Code—No ALT Tag: **<img src="examples/specttooltip.gif">.** **W**ith no ALT tag, screen reader says "Image," which leaves users wondering if they are missing anything important.

Accessible Summary ALT Tag

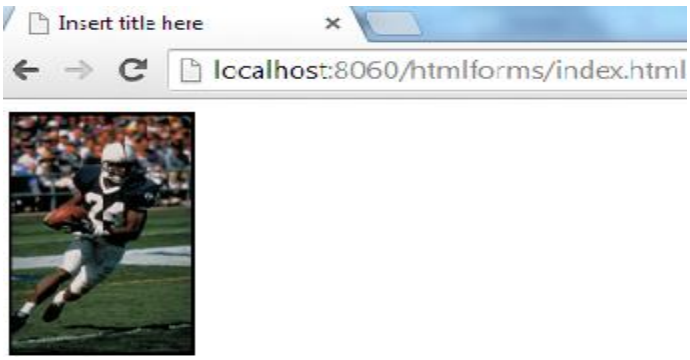
<...alt="saturated fat molecule" >

Less Accessible, Verbose ALT Tag

<...alt="image of straight saturated fat with 18 carbons" >

LONGDESC Attribute: This is an attribute that is hidden in visual browsers, but recognized by some screen readers. View the LONGDESC Code

<img src="example/runningbasketbalplayer.jpg" alt="photo of basketballplayer" width="102" height="154" longdesc="ball.html" >



2.1.5. HTML Language Tags

The LANG tag (i.e. the `lang=""` attribute) is designed to signal screen readers pronunciation engines to switch to another language.

Declaring a U.S. English Page : <html lang="en-US"> … </html>

Declaring a British English Page : <html lang="en-GB"> … **</html>**

Screen readers supporting this tag could switch to a British accent.

Declaring a French English Page: Screen readers supporting this tag could switch to a French accent.

<html lang="fr"> … **</html>**

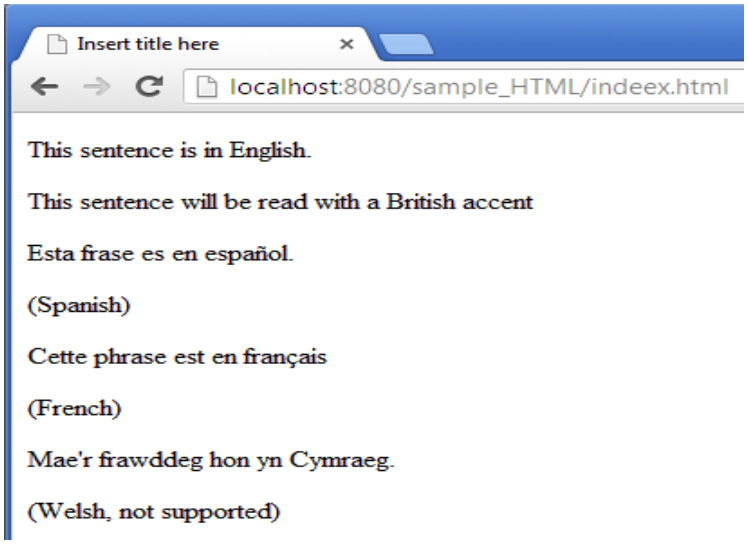
Switching Languages

If you switch languages within one page, you can embed the LANG attribute in other tags such as a P, TD, SPAN, DIV and other tags. For example

<u>View the Code : Test Text with Lang Tags</u>

1. ** This sentence is in English. **
2. ** This sentence will be read with a British accent **

3. Esta frase es en español. (Spanish)
4. Cette phrase est en français (French)
5. Mae'r frawddeg hon yn Cymraeg. (Welsh, not supported)



2.1.6. HTML Lists

Using Image Bullets (CSS): You can replace bullets with custom bullet images using the CSS attribute list-style-image:url(path). Example list with custom bullets: State's two colors are:

Blue White

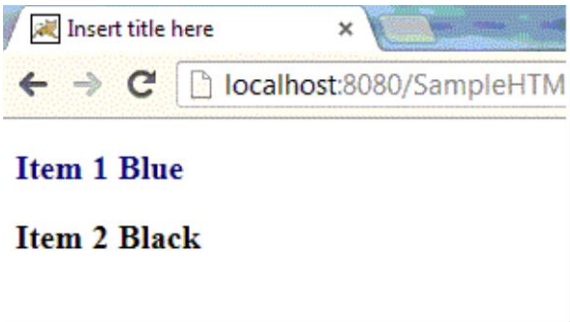
CSS file specification:

ol.paw {list-style-image:url(example/jaw.gif)}

View the HTML code:

<ol class="jaw"> <li>Blue</li> <li>White</li> </ol>

The list-style-image attribute replaces a bullet with the image. What a screen reader says: State's two colors are:



2.1.7. HTML Skip Navigation

It is recommended that a **"skip to content"** link be included before the site's main navigation tools. One link of this kind is usually sufficient. Users with screen readers can skip between major headers and sub-headers, which are tagged H2, H3, and H4.

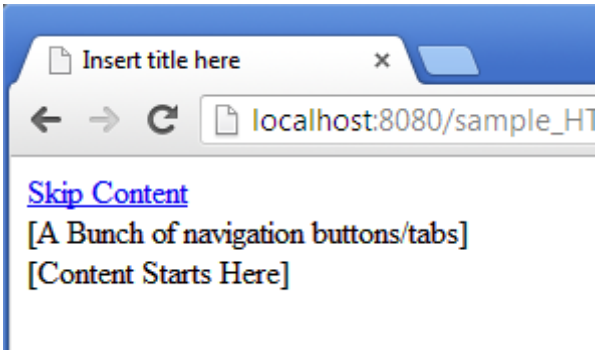
Skip Link Method 1—Hiding Text Off Screen

View the CSS

```
.offscreen {position:absolute; left:-10000px; top:auto; width:1px; height:1px; overflow:hidden; }
```

View the HTML

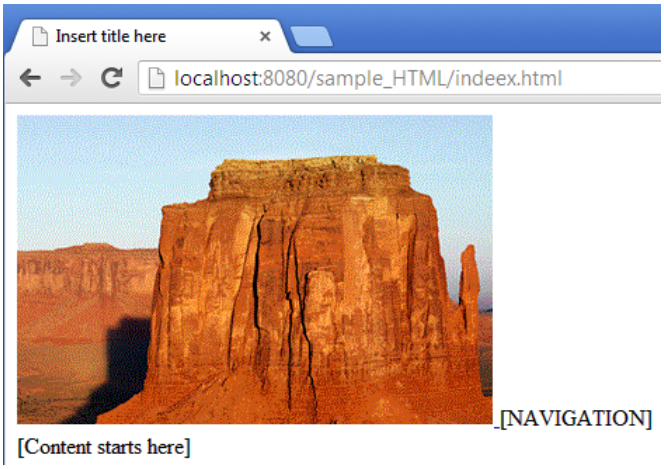
```
<a href="#skip" class="offscreen">Skip Content</a>

<!-- DIV tags are used to break the page into sections. No formatting is visible unless CSS formatting is applied-->

<div id="navigation">[A Bunch of navigation buttons/tabs]</div>

<div id="content">

<a name="skip" > </a > [Content Starts Here]

</div>
```



Skip Link Method 2—Pixel with ALT Tag

View the Code

```
<a href="#skip"> <img src="transparent-pixel.gif" alt="Skip Content " border="0" >

</a > [NAVIGATION]<div id="content"> <a name="skip" > </a > [Content starts here] </div>
```



2.1.8. HTML Tables and Titles for Web Pages

```
<table summary="Color names for black, white, red, blue, green, yellow in multiple languages">

<caption> Color names in multiple languages</caption>

<tr> <th> <scope="col"> Color </th> <th scope="col"> Spanish </th> <th scope="col"> French </th> <th scope="col"> Irish </th> th scope="col"> Welsh </th> </tr> </table>
```

How it works

```
<tr>...<td headers="spanish-romance-green">verde</td>...</tr>
```

```
<tr>...<th id="romance">Romance Language</th>...</tr>
<tr>...<th id="spanish" headers="romance">Spanish</th>...</tr>

<tr>...<th id="green">Green</th>...</tr>
```

```
<table summary="Example of a table that is coded to demonstrate accessibility"><caption>Example of an Accessible Data Table</caption> <thead> <tr><th scope="col">Student ID</th><th scope="col">FirstName</th> <th scope="col">Last Name</th></tr></thead>tr><th scope="row">91234</th><td>John</td><td>Jones</td></tr>
<tr><th scope="row">92345</th><td>Vicky</td><td>Brown</td></tr>
<tr><th scope="row">93456</th><td>Dan</td><td>Dixon</td></tr></table>
```

Student ID	First Name	Last Name
91234	John	Jones
92345	Vicky	Brown
93456	Dan	Dixon

2.2.Cascading Style Sheet Guidelines for Web Accessibility

Line Spacing and Margins: View the Code

body: {margin-left: 25px; margin-right: 25px; font-family: Verdana,Arial,Helvetica,sans-serif; line-spacing: 120%}

Without CSS : Text goes to edge of screen, is single-spaced and in Times New Roman (not ideal for monitors).

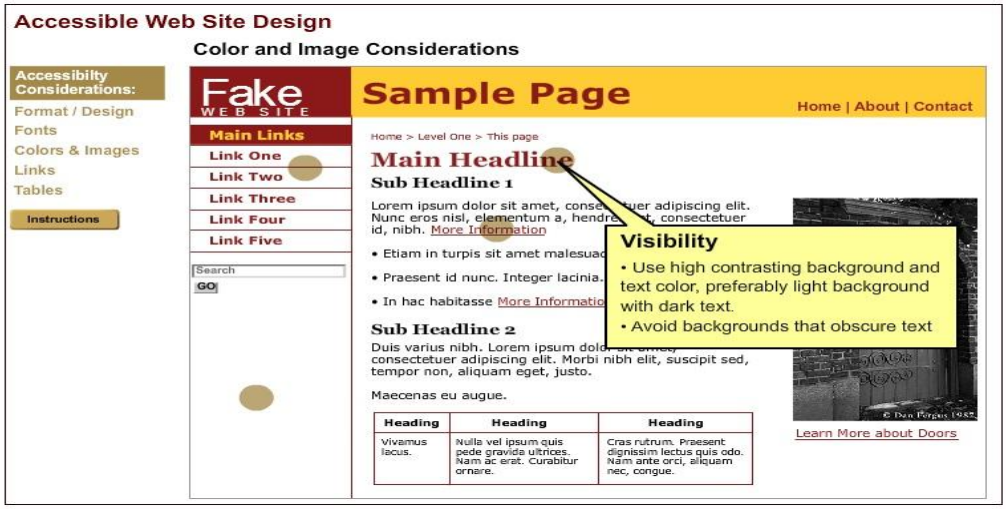
Coloring and Aligning H Tags

With CSS: h1 {color: #000066; text-align: center;} .. <h1 > Sample H1 </h1>

Without CSS : <h1 align="center"> <font color="#336699> Sample H1 </font> </h1>

Creating Formatting DIVs Resembling Tables

**.pinkdiv {border: 1px dotted black; background-color: #FFAADD; padding: 7px; margin-left: 150px; margin-right: 150px;}... I want a centered pink div area **



CSS Rollovers: Font Color Change: Links which turn Green

CSS File Specification

```
a {color: #000066; font-weight: bold} /*Navy Blue*/
a:hover, a:focus {color: #006633; font-weight: bold} /*Green*/
a:visited {color: #9900FF; font-weight: bold} /* Purple */
```

HTML Code : `<a href="#">Green Link </a>`

View the float:right Code : `.projectbox { float:right;width:45%;background-color:#CCF;color: black;padding:7px;margin: 5px;}`

CSS Rollovers: Changing the Background Color: `a {padding-left: 3px, padding-right: 3px} a:hover, a:focus {color: #white; background-color: #333333 padding-left: 3px, padding-right: 3px }`

CSS Rollovers: Classes for Special Navigational Links

```
/* .mainnav means the "mainnav" class*/
a.mainnav {font-weight:bold; font-size: 125%; color: #006633 ;text-decoration: none }
a:visited.mainnav {font-weight:bold; font-size: 125%; color: #006633; text-decoration: none }
/* You need to specify identical formats for a.button and a:visited.button in order to keep appearance the same*/
a:hover.mainnav, a:focus.mainnav {color: black; text-decoration:underline}
```

HTML Code: `<a href="#" class="mainnav" >Main Navigation Link </a >`

Image Rollovers: `p.prise a:hover, p.prise.focus {display:block; color: black;text-decoration: none; background-image:url(riseme.gif); padding: 5px; border: 1px solid black }`

CSS: Ensuring that Text Order is Logical : View the float:right Code

CSS: .projectbox {float:right;width:45%;background-color:#CCF;color: black;padding:7px;margin: 5px;}

HTML : <body> <h2>Projects can Include</h2>[ORDERED LIST]

CSS. Button Style Links: Simple

a.button1 {font-weight:bold; font-size: 125%; color: black;background-color: #FFCCFF;text-decoration: none;padding-left: 3px; padding-right: 3px}

a:visited.button1 {font-weight:bold; font-size: 125%; color: black;background-color: #FFCCFF;text-decoration: none;padding-left: 3px; padding-right: 3px}

2.3. JavaScript Guidelines for Web Accessibility

1. Trigger events with active input from users rather than nonactive triggers.
2. Provide a mechanism for the user to freeze or slow-down any moving or blinking objects, particularly those that contain text
3. Make scripts and applets keyboard operable
4. Provide a<NOSCRIPT>option for all scripts

```
<SCRIPTtype="text/tcl">...some Tcl script to show a billboard of sports scores... </SCRIPT> <NOSCRIPT> <P> To access today's scores <A href="scores.html">visit our text-only version.</A> </NOSCRIPT>
```

2.3.1. Drop-Down Navigation Selections

Drop-down select menus have often been used to redirect readers when they select an option from the drop-down menu. Here is an example of the misuse of a select menu:

```
<script type="text/javascript"><!--
function redir(sel){
document.location=sel.options[sel.options.selectedIndex].value;
} //--></script>
```

2.3.2. Form Validation

```
<script type=”text/javascript”><!–
function validateForm(){
if(myForm.fName == "" || myForm.lName == "" || myForm.fColor == ""){
alert("Please fill in all fields."); return false; }
return true; }
//--></script>
```

What would you like to drink with your sandwich?

Milk
Soda
Coffee
Tea

```
<form name="myForm" action="some-script.pl" method="post"
onsubmit="return validateForm(this)"><fieldset>
```

2.3.3. Inner HTML and Roll overs

```
<script type="text/javascript"><!--
onload = function(){var p = document.createElement("P");
….....
p.appendChild(document.createTextNode("Hi, this is what I
have to say!"));
document.body.appendChild(p); }
//--></script>
```

Roll-overs

```
<script type="text/javascript"><!--var img1 = [], img2 = [];
img1[0] = new Image(); img1[0].src =
"http://localhost/html.gif";
function roll(n,s){ document.images[n].src =
window[n][s].src;}
//--></script>
```

2.4.UI Design considerations for Accessible Web

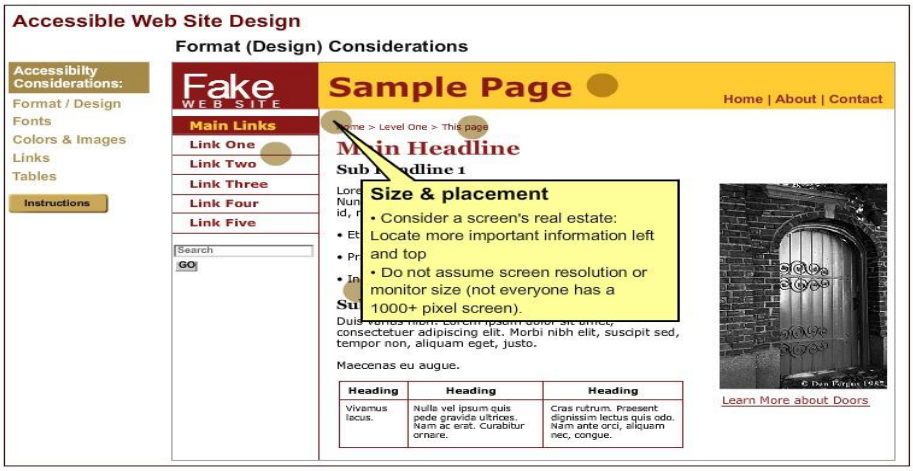
While designing the Web sites/pages, there are many factors that should seriously be considered to enable the 'Web Accessibility'. Few important factors are:

1. Format(Design) Considerations
2. Fonts
3. Colors
4. Images
5. Tables
6. Links

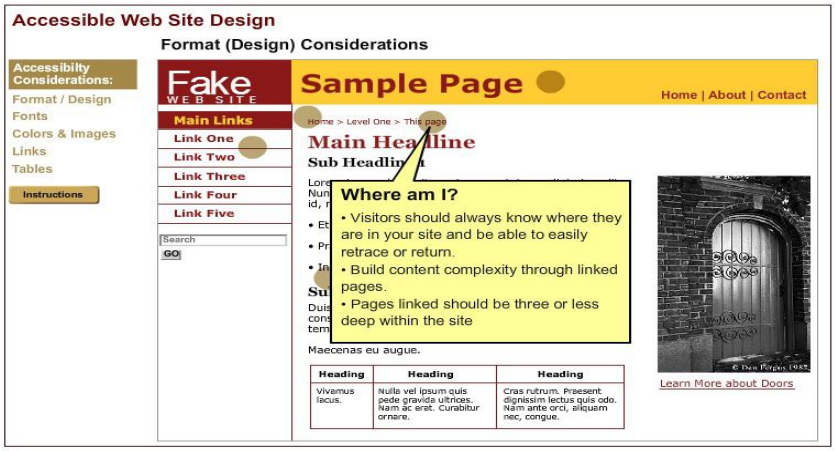
1. Format (Design) Considerations:

- Consider a screen's real estate: Locate more important information left and top. Do not assume screen resolution or monitor size Facilitate scanning: Most viewers (about 75%) first scan text and menus for information (and ignore visuals).Increase detail and complexity with linked pages (fewer than 20% read word-by-word)

- Standardize presentation and navigation :Locate logos, menus and features consistently and predictably longer pages "back to the top" Intuitively label links for content and page URL Reduce white space on main/menu pages that detail content indexes and facilitate searching

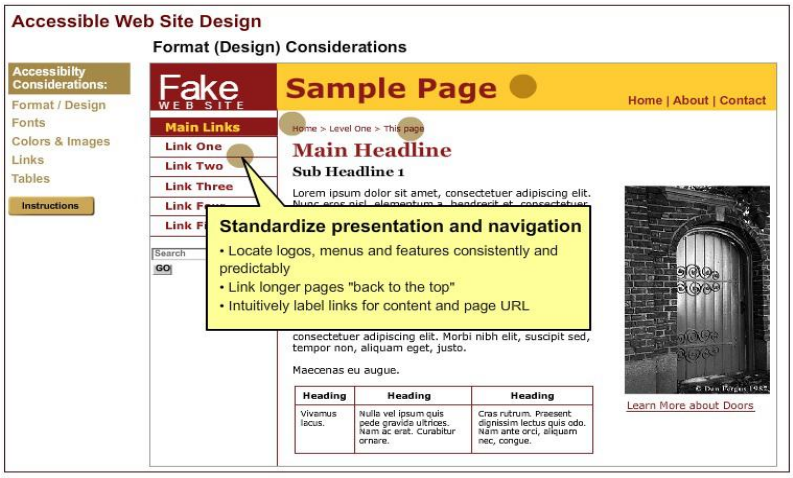


- Build content complexity through linked pages: Web-pages should be linked three or less deep within the site. Visitors should always know where they are in your site and be able to easily retrace or return
- Make text easier to read : Make liberal but consistent use of bullets, headings, sub-headings, and font size Separate blocks of text with white space .Do not fill the screen with text: indent, center blocks of text with margins left and right into columns

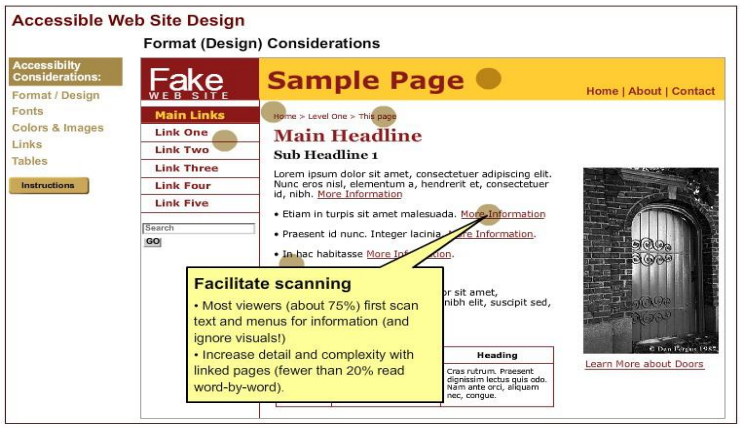


2. Font Considerations:

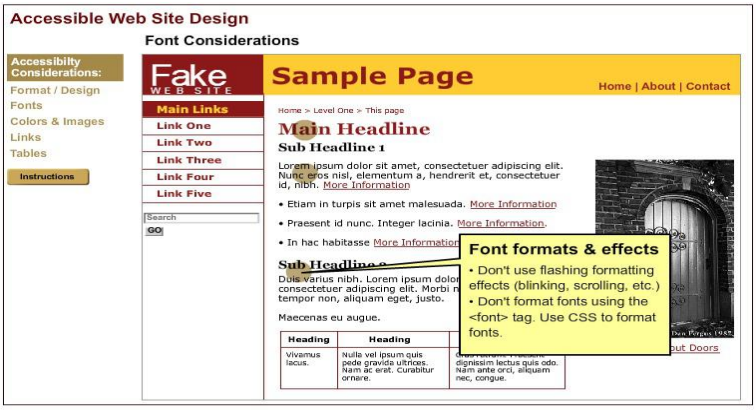
- **Fonts: Use standardized or common fonts if not using "default"**
  Font size display is affected by monitor size, screen resolution, as well as the browser's settings. Georgia (serif), Trebuchet and Veranda (sans serif) have been specifically designed for Web display; Times Roman (serif) and Helvetica and Ariel (sans serif) are also common. For maximum accessibility, font sizes should be coded in relative sizes rather than fixed or absolute sizes. If fixed:



- Minimum font size should be 10 (PC) or 12 (Macintosh) or medium or "3"
  Use a larger font size for sites attracting small children and seniors
  (Based on the Web Content Accessibility Guidelines (WCAG) published by the Web)

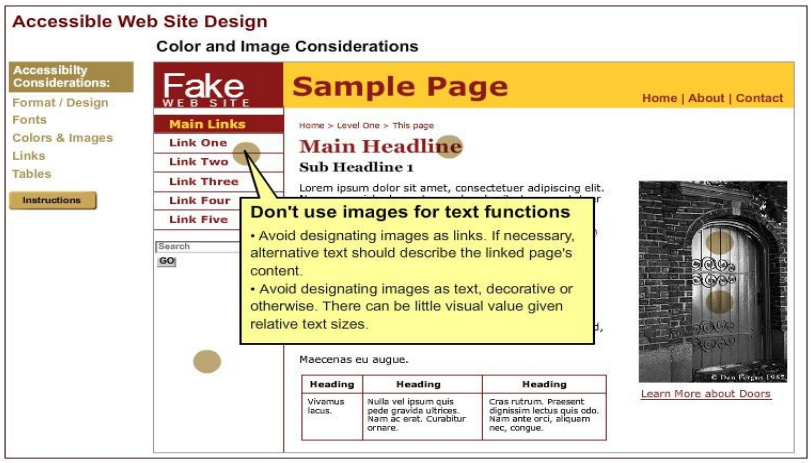


- Research shows little difference in reading speed or user preference between 10-point Times Roman, Georgia, Helvetica, or Verdana fonts. Use one font for identification, directions and navigation; use another for content Do not use more than two per Web site



3. Colors considerations: Use color judiciously

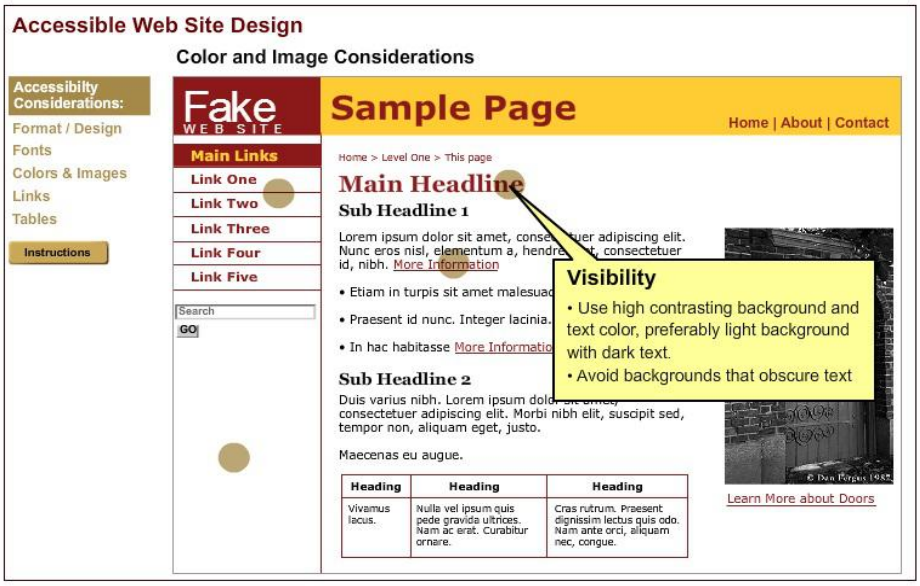
- Either specify all colors (BGCOLOR, TEXT, LINK, VLINK, and ALINK) or none "None" will default to the settings on your visitor's computer
  Specifying only some colors will let others default; your designated colors may be an ugly combination with these defaults
- Use high contrasting background and text color, preferably light background with dark text. Avoid backgrounds that obscure text .Use a browser safe palette of 216 colors. These colors are standardized for all browsers (IE, Netscape, Opera, Lynx) and platforms (PC and Macintosh). These are defined as "#RRGGBB" (Red, Green, Blue) paired values (00, 33, 66, 99, CC, and FF)



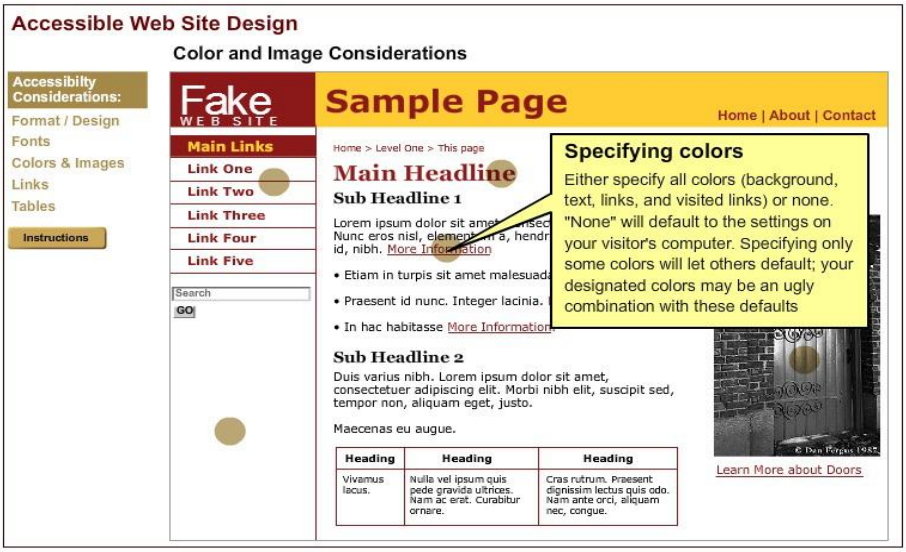
4. Images Considerations:

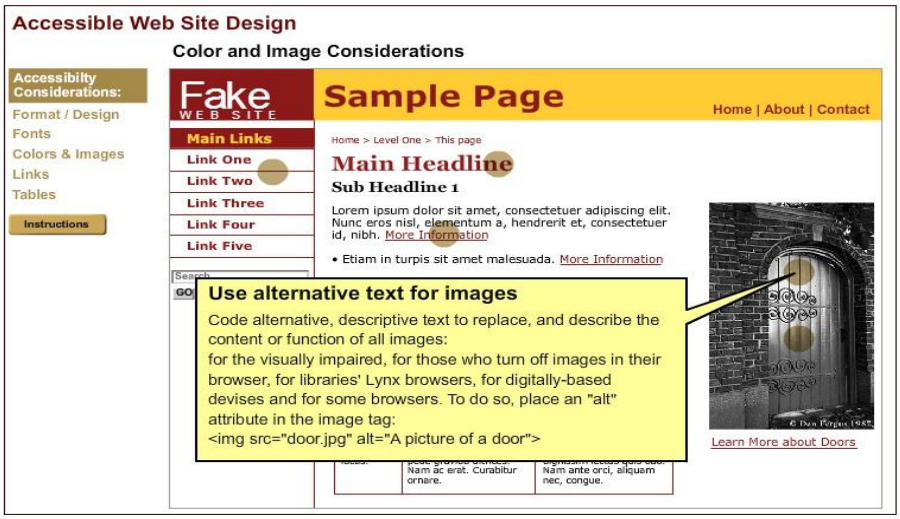
- Images should be no more than 75 pixels per inch. This is the conveyable limit on the Internet; any more results in slow downloads with no increase in resolution or visibility. Use .jpg files for images as photographs with shaded coloration; use .gif files for images as graphics with broad fixed color fields

- Code alternative, descriptive text to replace, and describe the content or function of all images: for the visually impaired, for those who turn off images in their browser, for libraries' Lynx browsers; for digitally-based devises; for 30% of all browsers



- In HTML code, <alt> designates the alternative text for an image.
<img src="logo.gif" alt="company's logo"> company's logo replaces the image
<img src="space.gif" alt="*"> or <img src="space.gif" alt="">A blank replaces this graphic place holder or decorative graphic

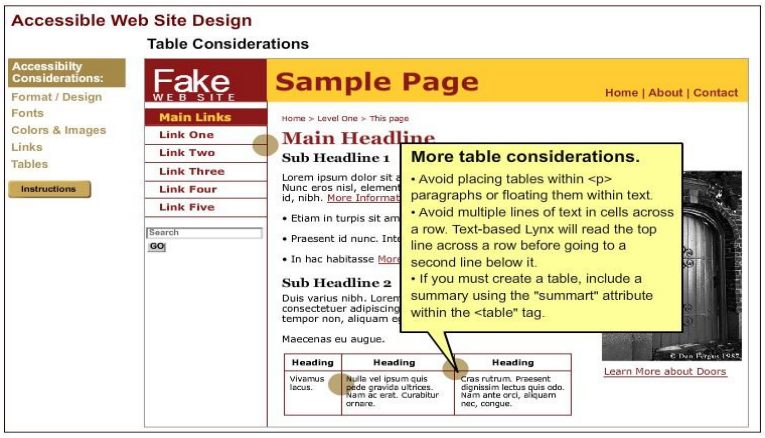




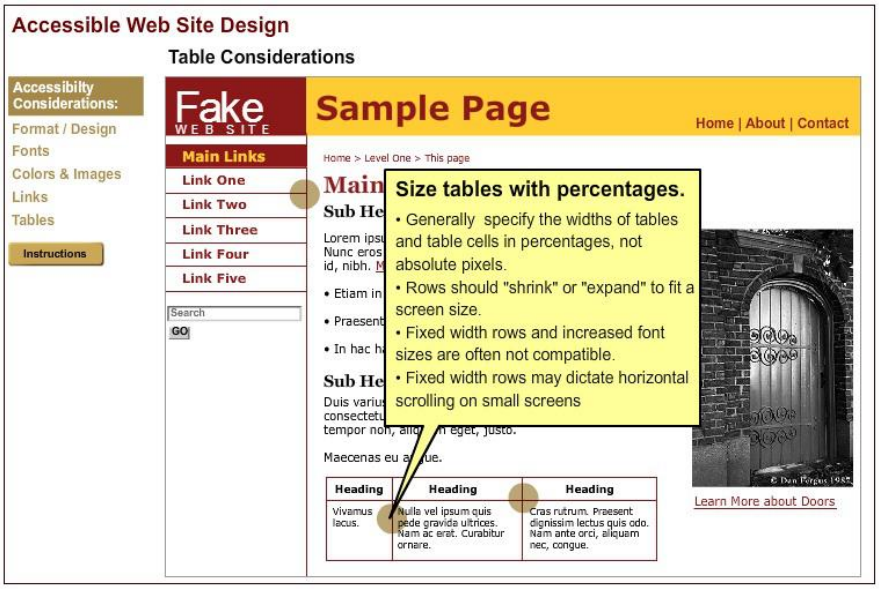
5. Tables Considerations:

Use tables conservatively to format presentation and design

- Generally specify the widths of tables and table cells in percentages, not absolute pixels. Generally rows should "shrink" or "expand" to fit a screen size.



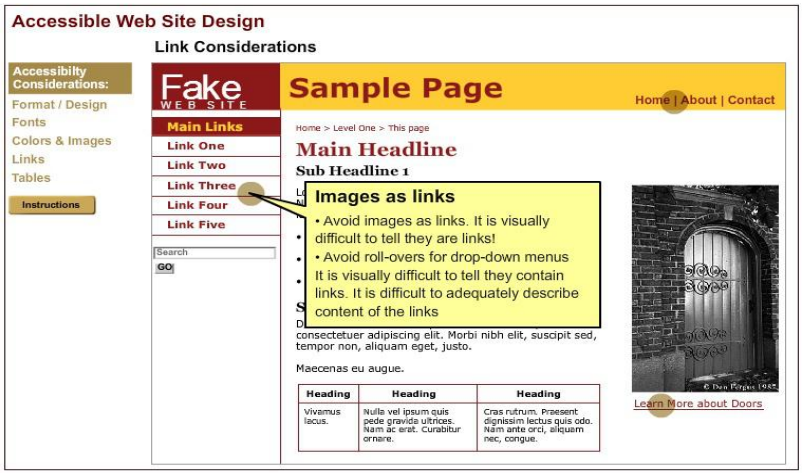
Fixed width rows and increased font sizes are often not compatible.
Fixed width rows may dictate horizontal scrolling on small screen. Avoid placing tables with <p> paragraphs or floating them with text. Avoid multiple lines of text in cells across a row.



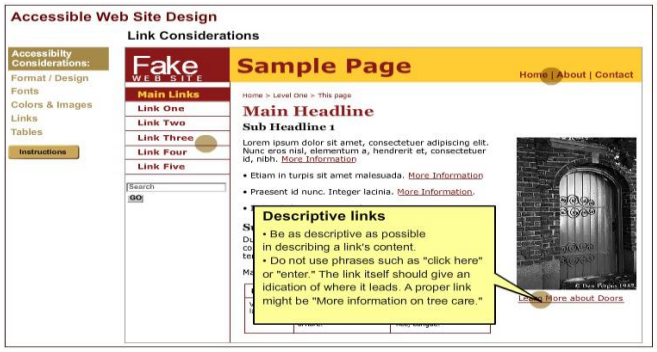
1. Links Considerations:

- Be as descriptive as possible in describing a link's content. Use a space | vertical bar | space to separate links which occur consecutively

- Do not use phrases such as "click here" "enter". Avoid images as links It is visually difficult to tell they are links!



- Avoid roll-overs for drop-down menus. It is visually difficult to tell they contain links. It is difficult to adequately describe content of the links



3. References

To get more details on Web Accessibility, the following references will help:

1. W3C Web Accessibility Initiative : http://www.w3.org/WAI/
2. W3C **Web Content Accessibility Guidelines version 2.0** (WCAG 2.0) http://www.w3.org/TR/WCAG/
3. US federal government agencies are required by law to make their websites comply with the requirements in **Section 508** : http://www.section508.gov/
4. Indian Government has accessibility guidelines for Indian government websites : http://web.guidelines.gov.in/#&panel1-1
5. Knowmax page: https://knowmax.ultimatix.net/sites/Accessibility_CoE-cop/default.aspx
6. Knome: https://knome.ultimatix.net/communities/4926-tcs-accessibility-coe